

Zhengyang Yu

Curriculum Vitae

San Diego & Bay Area

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Education

2025 – Jun 2028 **Bachelor of Science in Computer Science**, *University of California, San Diego*, San Diego, CA

Expected Graduation: June 2028

Relevant Coursework: Advanced Data Structures, Design and Analysis of Algorithms, Computer Organization, Systems Programming, Machine Learning

2023 – 2025 **Associate of Science in Computer Science**, *Laney College*, Oakland, CA

Projects

2026 **Quotaquarium**, *Personal Project*

- Designed and built standalone C++ firmware for an ESP32-S3 device that visualizes live multi-account quota data through an animated, touch-controlled interface with configurable alerts.
- Implemented concurrent FreeRTOS tasks, adaptive HTTPS polling, and rate-limit-aware scheduling; optimized RGB565 rendering in PSRAM and added bounded logging and runtime diagnostics for reliable debugging.

2025 **HTML to ePub Converter**, *Personal Project*

- Engineered a Python document-processing pipeline with modular parsing, metadata embedding, and structured error handling for reliable HTML-to-ePub conversion.

2024 **Home Assistant Presence Plugin**, *Personal Project*

- Built a Python Home Assistant integration that derives real-time room occupancy from entry and exit sensor events.
- Designed an event-driven REST API pipeline with explicit state transitions for reliable smart-lighting automation and easier debugging.

Skills

Programming C++, C, Python, Java

Systems ESP32-S3, FreeRTOS, Linux, concurrent and event-driven systems, HTTPS, REST APIs, networking fundamentals

Engineering Object-Oriented Design, Data Structures, Algorithms, performance optimization, software testing, complex debugging

Tools PlatformIO, Git, Docker, VS Code, structured logging and runtime diagnostics

AI Tooling Codex-assisted implementation, refactoring, debugging, code validation, and technical documentation

Activities

2024–2025 Computer Consultant, Oakland Public Library - Elmhurst Branch

Supported community members with computer troubleshooting and taught basic software skills.

2025–Present Member, UCSD Envision Makerspace

Prototype embedded and electronics projects using makerspace fabrication tools.